

Impact of RAG on Medical QA in a Primary Care Context

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Introduction and Background

- **Current Difficulties in Medical QA:**
 - LLMs hallucinate,
 - Medical knowledge evolves
 - Mistakes are more costly than in other fields
- **Focus on Primary Care:**
 - First point of contact in healthcare
 - AI assistance can help alleviate the workload
- **Background Work:**
 - LLMs such as PMC-LLaMA are fine-tuned to the medical field and perform competitively with larger models
 - MedRAG is a toolkit designed to assist in implementing RAG into LLMs within a medical context
- **Main Question:**
 - Does the ability to obtain medical information (RAG) help an LLM reach better answers?

Key Tools and Models

BioMistral-7B: A Mistral-7B variant pre-trained on textual data from PubMed Central Open Access. It is used as the base model for all of our experimental setups due to its domain relevance and competitive performance against other LLMs. Biomistral-7B is loaded in 4-bit quantization to be used within a Google Colab environment. PMC-LLaMA was our initial LLM of choice, but was discarded due to running out of recursion depth.

FAISS: A library for efficient similarity search and clustering of dense vectors. Assists in faster RAG performance.

MedCPT: A biomedical IR model that includes a pair of retriever and re-ranker closely integrated by contrastive learning. MedCPT was trained using 255M query-article pairs and is a publicly available resource.

StatPearls: A lighter-weight corpus from NCBI. Covers many medical topics and serves as the extra information that the LLM can retrieve information from.

Dataset – MedMCQA

- An open source dataset containing multiple choice questions designed to address real-world medical entrance exam questions.
- Subjects included in our dataset include: "Medicine", "Pharmacology", "Microbiology", "Social & Preventive Medicine", "ENT", "Psychiatry", "Orthopaedics", and "Skin".
- An 80/20 train/test split was chosen for our study.

```
{  
  "question": "A 40-year-old man presents with 5 days of prod  
  "exp": "Circulating levels of leukocytes and their precurs  
  "cop": 1,  
  "opa": "Leukemoid reaction",  
  "opb": "Leukopenia",  
  "opc": "Myeloid metaplasia",  
  "opd": "Neutrophilia",  
  "subject_name": "Pathology",  
  "topic_name": "Basic Concepts and Vascular changes of Acute  
  "id": "4e1715fe-0bc3-494e-b6eb-2d4617245aef",  
  "choice_type": "single"  
}
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RAG vs. RAFT

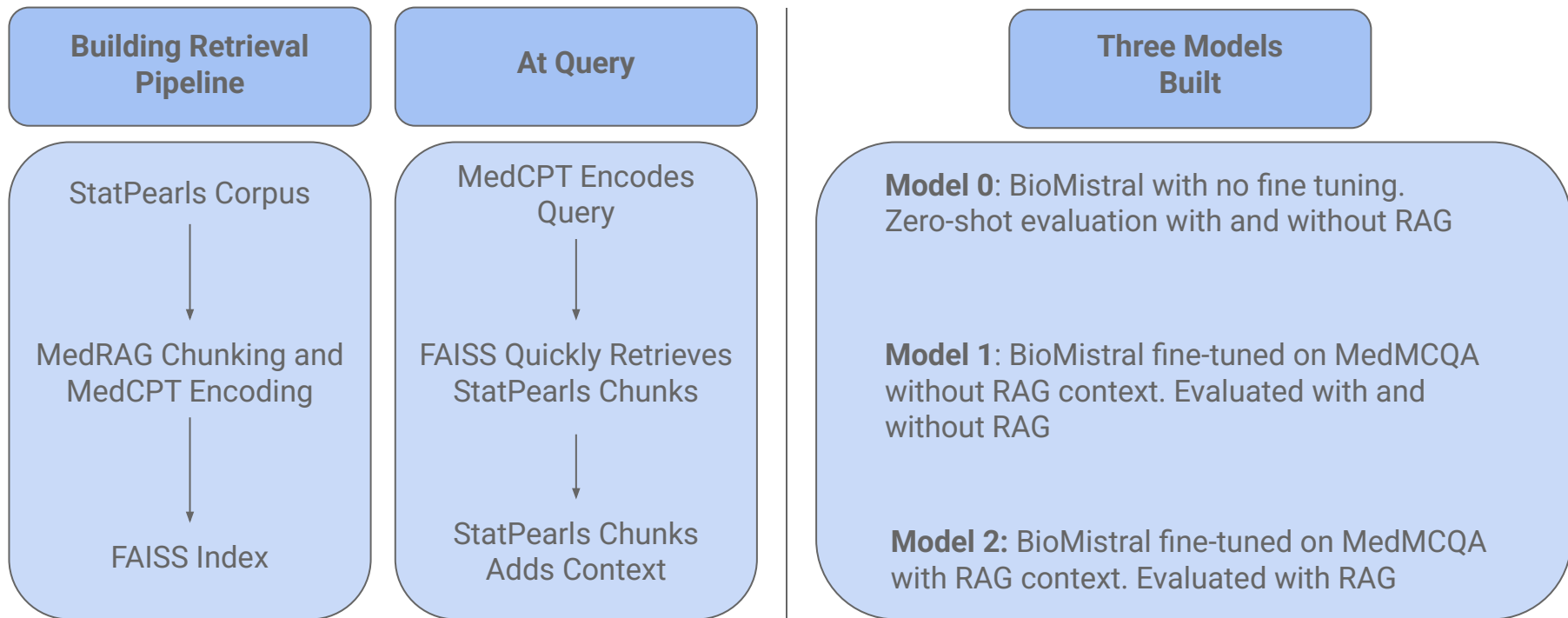
Retrieval Augmented Generation (RAG):

- RAG retrieves relevant information from a document index at inference time and uses said information as context when generating a response.
- Good at providing additional information to better inform the LLM's response, but may be limited in its ability to utilize the extra information.

Retrieval Augmented Fine Tuning (RAFT):

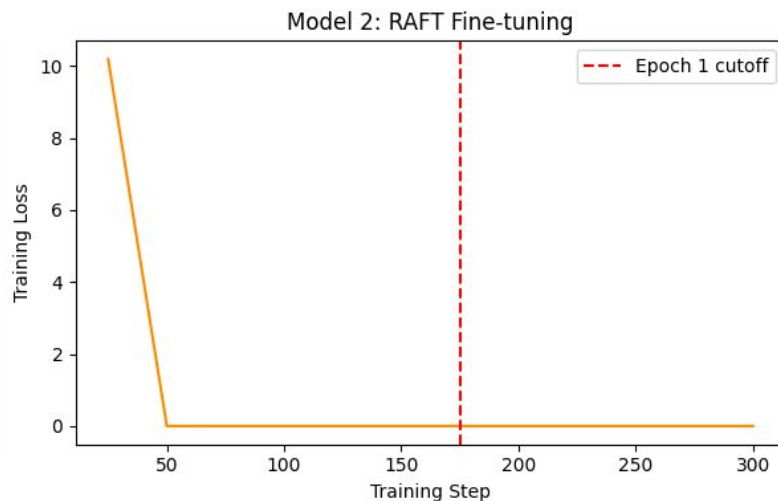
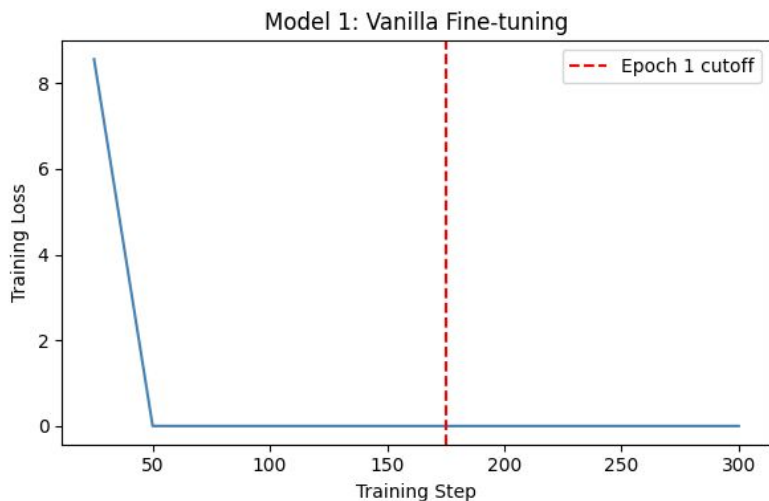
- RAFT is built off of the weakness mentioned in RAG. RAFT is very similar in structure to RAG, but RAFT is trained to better utilize the retrieved information.

System Architecture and Experimental Design



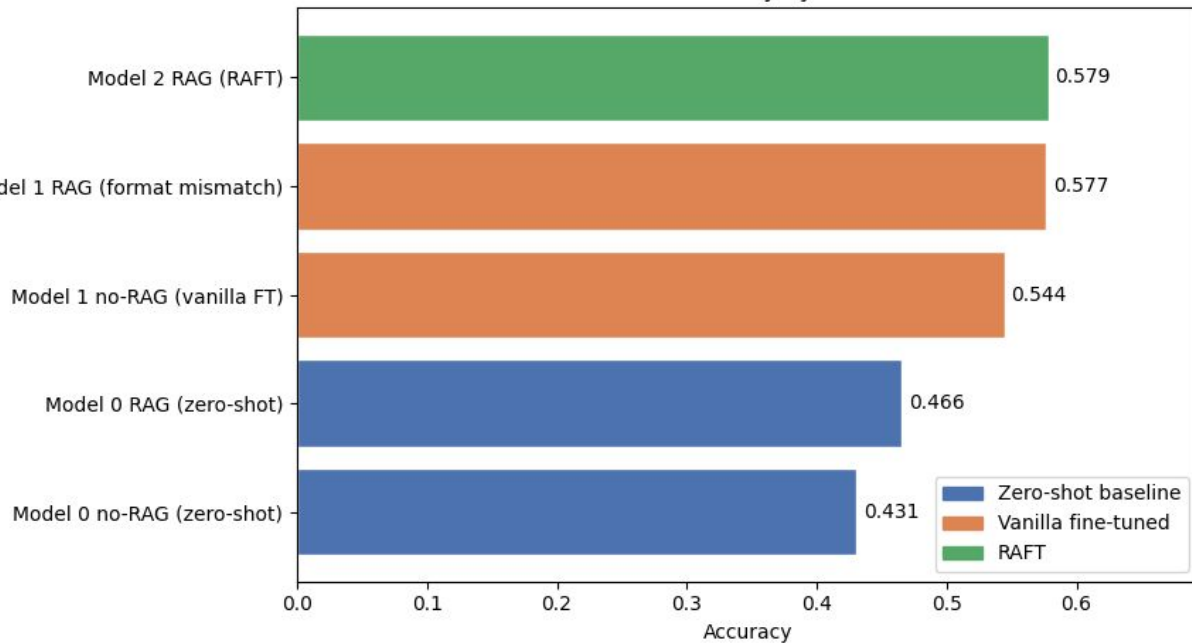
Results

- There was an early sudden drop off in training loss - likely due to the model getting a better sense of the format
- MedMCQA does not give clear reasons or logic for answers which could be the reason for the failure to continue to improve



Results

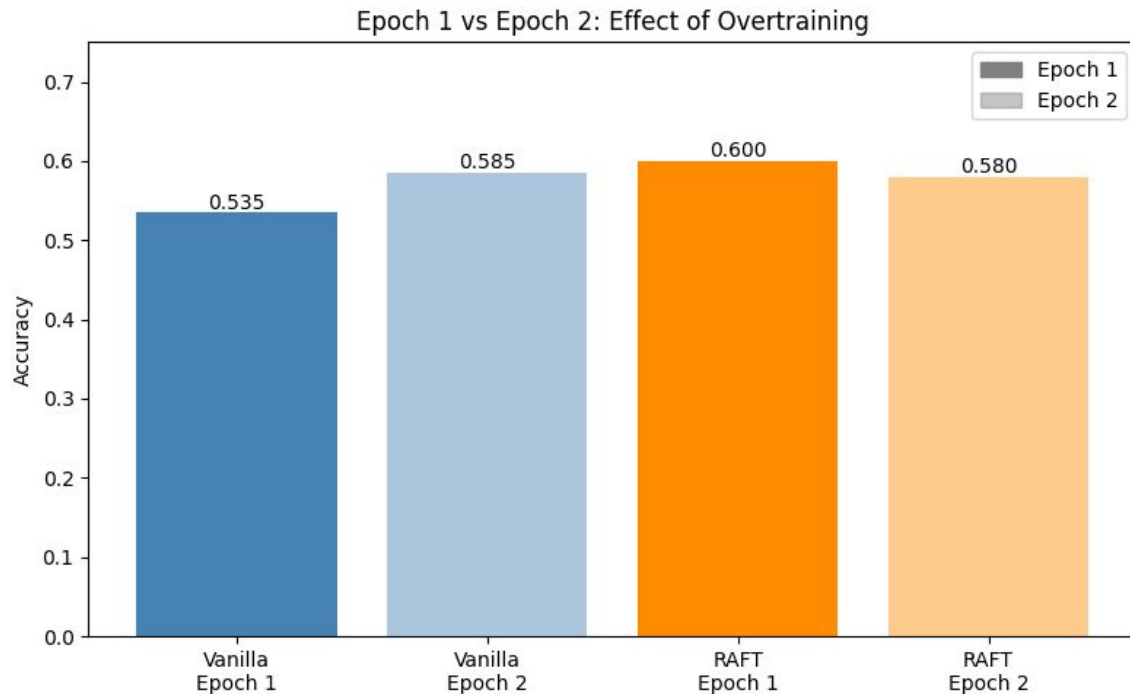
MedRAG Accuracy by Condition



- The model did improve with both the introduction of RAG and fine tuning
- Performance boost through RAFT was unremarkable

Results

- Because from the rapid drop on loss on training and subsequent stagnation, we assumed that we had overtrained the models
- We thus evaluated performance on Epoch 1 weights.
- We compared Epoch 1 to Epoch 2 performance to determine whether performance worsened
- Interestingly, while the RAFT model was overtrained, the more heavily trained Vanilla model would have performed better



Discussion

Model could have been improved with:

1. Retrieval improvements - some of the failures were as a result of semantically similar but irrelevant RAG texts being retrieved. CoT prior to the retrieval may allow for more relevant search
 - a. For example a question asking for a categorization of a bat bite to a child's face, chunks from stat pearls about flea bits and human bites were pulled rather than bite categorization information
2. CoT reasoning to reason through an answer could have also improved accuracy rather than just correctly inputting the correct token
3. We were limited by the availability of GPU Ram so could not bring in this context in a reasonable time. Fine tuning on big datasets takes a lot of compute!
4. It would have also been useful to compare few shot performance with an non-fine tuned model against the fine tuned performance to see if the fine tuning merely taught the model the format of the answers

Conclusions

- We build a MedRAG enabled Primary Care Focused QA
- We compared RAG using MedCPT retriever to collect chunks from the StatPearls dataset
- We looked at 5 different arms comparing raw BioMistral-7B, Fine-Tuned with MedMCQA and RAG Augmented Fine Tuning using MedMCQA without RAG and with RAG.
- Results were generally modest with improvements from RAG and Fine tuning.
- RAFT did not have significant improvements
- The model could likely be improved by using CoT reasoning before RAG to ensure more relevant retrieval occurred.